Fiscal Year:	FY 2011	**Task Last Updated:**	FY 12/08/2011
PI Name:	Lathan, Corinna E. Ph.D.		
Project Title:	Mental Representation of Spatial Cues (3D-Space)		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	clathan@atinc.com	**Fax:**	FY 301-585-9075
PI Organization Type:	INDUSTRY	**Phone:**	301-495-0770
Organization Name:	AnthroTronix, Inc.		
PI Address 1:	8737 Colesville Rd. L-203		
PI Address 2:			
PI Web Page:	www.atinc.com		
City:	Silver Spring	**State:**	MD
Zip Code:	20910	**Congressional District:**	8
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	ILSRA 2001
Start Date:	05/02/2005	**End Date:**	09/30/2011
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Leveton, Lauren	**Contact Phone:**	
Contact Email:	lauren.b.leveton@nasa5.gov		
Flight Program:	ISS		
Flight Assignment:	ISS Co-Investigator to ESA managed flight study. NOTE: Received NCE until 9/30/2011 per C. Guidry/JSC (10/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Clement, Gilles (Centre National de la Recherche Scientifique (CNRS))		
Grant/Contract No.:	NNJ05HD92A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Accurate perception and localization of objects in the environment are prerequisites for spatial orientation and reliable performance of motor tasks. Humans have mental representations of their environment based on sensory information and experience. Previous space studies carried out by our group have shown that cognitive functions such as mental rotation and the internal representation of the vertical axis of three-dimensional (3D) objects and written characters were impaired during and after spaceflight. The absence of the gravitational frame of reference during spaceflight could be responsible for these disturbances. The 3DSPACE experiment is designed to allow further identification of the role of gravity in the mental representation of 3D objects and environments. This experiment includes motor tests complemented by psychophysical measurements, in order to distinguish the effects of cognitive versus perceptual-motor changes due to microgravity exposure. See also https://
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this study could have important implications for human performance during space flight, particularly with respect to tasks involving distance estimation, such as docking and teleoperation of robotic assets. There may also be consequences for a space habitat design if squared volumes do not look square to astronauts. Virtual reality training may be a way to train the astronauts to compensate for such altered spatial representation with the goal of developing countermeasures to alleviate any associated performance risks.
Task Progress:	Summary of Work--Fall 2010 During the current reporting period (May – September, 2010), data were collected for five subjects. Inflight and Postflight data were collected for Subject 6. Preflight data were collected for Subjects 7 and 8 as well as two backup Subjects. We also completed postflight debriefs for 4 additional subjects. Also, during the current period of performance, investigators supported both preflight and postflight data collections at multiple locations and conducted analysis of all available data. Administrative tasks conducted during the current reporting period included team meetings and teleconferences with the ESA PI, progress documentation and reporting, travel to and from meetings and data collection sites, and coordination of the training and BDC with NASA and ESA personnel. SUPPLEMENTAL REPORTING FOR FINAL REPORT--DECEMBER 2011 Over a three-year period, pre-, in-, and post- flight data were collected on eight International Space Station Crew Members. The results include the following: 1) A 3D cube with smaller height than width is perceived as a perfect cube in-flight; 2) Hand drawn objects in-flight have greater height and smaller width than on the ground. Words written vertically in-flight are shorter (total vertical length of the word) than pre-flight, but longer immediately after return; and 3) Vertical distances are overestimated on Earth, but not late in-flight, and shorter distances ranging from 100-1000 feet, are underestimated.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
Abstracts for Journals and Proceedings	Lathan CE, Skinner A, Clement G. "Mental representation of spatial cues during spaceflight (3DSPACE)." 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. , Apr-2011
Abstracts for Journals and Proceedings	Lathan CE, Clement GR, Skinner AD. "Mental representation of spatial cues (3DSPACE)." 2010 NASA Human Research Program Investigators' Workshop, Houston, TX, February 3-5, 2010. 2010 NASA Human Research Program Investigators' Workshop, Houston, TX, February 3-5, 2010. Abstract #1115. , Feb-2010
Abstracts for Journals and Proceedings	Lathan C, Skinner A, Clement G. "Mental Representation of Spatial Cues During Spaceflight - an Update on the 3DSPACE Flight Experiment." 2009 NASA Human Research Program Investigators' Workshop, League City, TX, February 2-4, 2009. 2009 NASA Human Research Program Investigators' Workshop, League City, TX, February 2-4, 2009. Abstract Number 1144. , Feb-2009
Abstracts for Journals and Proceedings	Lathan C, Clement G, Lockerd A. "Mental representation of spatial cues during spaceflight (3DSPACE)." 2008 NASA Human Research Program Investigators' Workshop, League City, TX, February 4-6, 2008. 2008 NASA Human Research Program Investigators' Workshop, League City, TX, February 4-6, 2008. , Feb-2008
Abstracts for Journals and Proceedings	Clement G, Lathan CE, Lockerd A. "Mental representation of spatial cues during spaceflight (3DSPACE)." Life in Space for Life on Earth. 29th Annual ISGP Meeting, 10th ESA Life Sciences Symposium, 24th Annual ASGSB Meeting, ELGRA Symposium, Angers, France, June 22-27, 2008. Life in Space for Life on Earth. 29th Annual ISGP Meeting, 10th ESA Life Sciences Symposium, 24th Annual ASGSB Meeting, ELGRA Symposium, Angers, France, June 22-27, 2008. , Jun-2008
Abstracts for Journals and Proceedings	Clement G. "Perception of depth and distance in microgravity and in vestibular patients." 80th Annual Meeting, Los Angeles, CA, May 3-7, 2010. Aviation, Space, and Environmental Medicine 2009 Mar;80(3):243. http://www.ingentaconnect.com/content/asma/asem/2009/00000080/00000003 ;jsessionid=33994sce9i59d.alexandra , Mar-2009
Articles in Other Journals or Periodicals	Elmann-Larsen B (Ed.); Clement G. "3DSPACE: How gravity influences the accuracy of our perception of dimension and depth." ESA Human Spaceflight Science Newsletter. February 2009, p. 12-15. http://wsn.spaceflight.esa.int/docs/HumanSpaceflightScienceNewsletters/2009/Newsletter%202%20-%20February%202009.pdf , Feb-2009
Articles in Other Journals or Periodicals	Elmann-Larsen B (Ed.); Clement G. "Weightlessness is tricking our senses – ESA's three separate ISS experiments in neurosciences reveal some of the secrets." ESA Human Spaceflight Science Newsletter. October 2011, p. 11-13. , Oct-2011
Articles in Peer-reviewed Journals	Clement G, Fraysse MJ, Deguine O. "Mental representation of space in vestibular patients with otolithic or rotatory vertigo." Neuroreport. 2009 Mar 25;20(5):457-61. PubMed PMID: 19297738 , Mar-2009

Articles in Peer-reviewed Journals	Clement G, Lathan CE, Lockerd A, Lorigny E. "Mental representation of spatial cues during spaceflight (3D-Space)." Journal of Gravitational Physiology 2008;15:15-6. , May-2008
Articles in Peer-reviewed Journals	Clément G, Skinner A. Lathan C. "Distance and size perception in astronauts during long-duration spaceflight. " Life. 2013;3(4):524-37. https://doi.org/10.3390/life3040524 ; PubMed PMID: 25369884; PubMed Central PMCID: PMC4187133 , Dec-2013
Books/Book Chapters	Clement G, Reschke MF. "Neuroscience in Space." New York : Springer, 2008. ISBN: 9780387789491, Aug-2008
Books/Book Chapters	Clement G. "Fundamentals of space medicine. 2nd edition." New York : Springer, 2011. ISBN: 9781441999047, Jun-2011
Papers from Meeting Proceedings	Clement G, Lockerd A, Lathan C. "Perception of depth in microgravity during parabolic flight." 16th IAA Humans in Space Symposium, Beijing, China, May 20-24, 2007. 16th IAA Humans in Space Symposium, Beijing, China, May 20-24, 2007. , May-2007